



# Applications of List Decoding to Tracing Traitors

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**EE605 Project Presentation**

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## Introduction - What are traitors?

An important problem in the protection of intellectual property is how to deter piracy. Hence, it is crucial to design efficient **trace-ability (TA)** schemes that can reliably identify guilty users.

Traitors refer to users who collude to create unauthorized copies of content. The techniques ahead enable the construction of fast tracing algorithms that can reliably identify at least one guilty user, and even extend to finding all possible colluding groups. An example where these schemes could be applied is a movie where different copies have segments filmed from different camera angles.

## Algebiac-Geometry codes

Suppose  $X$  is a smooth curve over a finite field  $\mathbb{F}_q$ , and you pick  $r$  distinct rational points  $\mathcal{P} = \{P_1, \dots, P_r\}$  on it, along with another point  $P_0$  not in  $\mathcal{P}$ . Fix an integer  $\ell$ . Then the one-point AG code is formed by evaluating functions  $f$  at the points  $P_1, \dots, P_r$ , where  $f$  is allowed to have poles only at  $P_0$ , with multiplicity at most  $\ell$ , giving codewords of the form  $(f(P_1), \dots, f(P_r))$ . When  $2g - 2 < \ell < r$ , the code has dimension  $\ell + 1 - g$  and minimum distance at least  $r - \ell$ .

Reed-Solomon codes arise as a special case when the curve is the projective line, the points  $P_i$  correspond to field elements,  $P_0$  is the point at infinity, and  $\ell = k - 1$ , where  $k$  is the dimension.

## Background - Definitions, Lemmas and Theorems (1)

### Descendants of a Coalition

If  $C$  is a code of length  $r$  on a (finite) alphabet  $Q$ , we write the codewords as  $x = (x^{(1)}, \dots, x^{(r)})$ , where  $x^{(i)} \in Q$  for  $1 \leq i \leq r$ . Subsets of  $C$  will be called *coalitions*. For any coalition  $C_0 \subseteq C$ , we define the set of *descendants* of  $C_0$ , denoted  $\text{desc}(C_0)$ , by

$$\text{desc}(C_0) = \{w \in Q^r : w^{(i)} \in \{x^{(i)} : x \in C_0\}, 1 \leq i \leq r\}$$

## Background - Definitions, Lemmas and Theorems (2)

### Definition (1)

A code  $C$  is a  $c$ -TA code if for all coalitions  $C_i$  of size at most  $c$ , if  $w \in \text{desc}(C_i)$  then there exists  $x \in C_i$  such that  $|I(x, w)| > |I(z, w)|$  for all  $z \in C \setminus C_i$ .

### Definition (2)

A code  $C$  is a  $c$ -IPP code if for all  $w \in \text{desc}_c(C)$ , the intersection of the coalitions  $C_i$  of size at most  $c$  such that  $w \in \text{desc}(C_i)$  is nonempty.

### Lemma (1)

Every  $c$ -TA code is a  $c$ -IPP code.

## Background - Definitions, Lemmas and Theorems (3)

### Theorem (1)

*Suppose  $C$  is a code of length  $r$ ,  $c$  is a positive integer, and the minimum distance  $d$  of  $C$  satisfies*

$$d > r - \frac{r}{c^2}.$$

*Then*

1.  $C$  is a  $c$ -TA code;
2. if  $C_0$  is a coalition of size at most  $c$ , and  $w \in \text{desc}(C_0)$ , then
  - 2.1 there exists an element of  $C_0$  within distance  $r - \frac{r}{c}$  of  $w$ , and
  - 2.2 every codeword within distance  $r - \frac{r}{c}$  of  $w$  is in the coalition  $C_0$ .

## Background - Definitions, Lemmas and Theorems (4)

### Proof.

Let  $C_0$  and  $w$  be as in the statement of the theorem. By the pigeonhole principle, there exists  $x \in C_0$  such that  $|I(x, w)| \geq \frac{r}{c}$ , and so the distance from  $x$  to  $w$  is at most  $r - \frac{r}{c}$ .

Now suppose  $z \in C \setminus C_0$ . Then since  $w \in \text{desc}(C_0)$  and any two distinct codewords of  $C$  share fewer than  $\frac{r}{c^2}$  coordinates, we have

$$|I(z, w)| \leq \sum_{y \in C_0} |I(z, y)| < c \cdot \frac{r}{c^2} = \frac{r}{c}.$$

Thus,  $z$  has distance greater than  $r - \frac{r}{c}$  from  $w$ , and by definition,  $C$  is a  $c$ -TA code. □

## Utilizing List Decoding Principles (1)

### Theorem (2)

Let  $C$  be a Reed–Solomon code of length  $r$  and dimension  $k$  over a finite field  $\mathbb{F}_q$  of size at most  $r$ . If  $c$  is an integer,  $c \geq 2$ , and  $r > c^2(k - 1)$ , then  $C$  is a  $c$ -TA code and there is a traitor tracing algorithm that runs in time  $O(r^{15})$ . If  $r = (1 + \delta)c^2(k - 1)$  then the algorithm runs in time  $O(\frac{r^3}{\delta^6})$ . For  $r = \Theta(c^2k)$ , the run time is  $O(c^{30} \log_q^{15} N)$ .

## Utilizing List Decoding Principles (2)

### Theorem (3)

Let  $X$  be a nonsingular plane curve of genus  $g$  defined over a finite field  $\mathbb{F}_q$ ,  $\mathcal{P}$  a set of  $r$  distinct  $\mathbb{F}_q$ -rational points on  $X$ ,  $P_0$  an  $\mathbb{F}_q$ -rational point on  $X$  that is not in  $\mathcal{P}$ , and  $k$  an integer such that  $k > g - 1$ . Let  $c$  be an integer such that  $c \geq 2$  and  $r > c^2(k + g - 1)$ , assume that  $q \leq 2^r$ , and assume the preprocessing described in [11] has occurred. Then the one-point AG code

$$C_X(\mathcal{P}, (k + g - 1)P_0)$$

is a  $c$ -TA code with a traitor tracing algorithm that runs in time polynomial in  $r$ .

## Utilizing List Decoding Principles (3)

### Theorem (4)

If  $k$  and  $c$  are positive integers,  $q$  is a prime power,  $q > c^2 \geq 4$ , and  $\delta$  is a real number such that

$$0 < \delta \leq \frac{q/c^2 - 1}{q - 1},$$

then there exists an explicit linear  $c$ -TA code over the field  $\mathbb{F}_q$  of length

$$r = O\left(\frac{k^2}{\delta^3 \log(1/\delta)}\right) \quad (\text{or length } r = \Theta\left(\frac{k}{\delta^2 \log^2(1/\delta)}\right))$$

and dimension  $k$  with a polynomial (in  $r$ ) traitor tracing algorithm.

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## The Relaxed Descendant Set

Recall the definition of a descendant of a coalition:

$$\text{desc}(C_0) = \{w \in Q^r : w^{(i)} \in \{x^{(i)} : x \in C_0\}, 1 \leq i \leq r\}$$

This definition required the  $i^{\text{th}}$  coordinate of the pirate  $w$  to be in the set of the  $i^{\text{th}}$  co-ordinates of the coalition members. The above definition treated  $w$  and  $x$  as *codewords*, or vectors in  $Q^r$ .

We generalize this notion to get a “relaxed” notion of TA and IPP by considering unordered sets of size  $r$  instead of  $r$ -length vectors.

Hence, we define TA and IPP *set systems* similar to TA and IPP *codes*, except that a pirate is an unordered set of  $r$  elements such that each element of the pirate is in some coalition member’s set.

## TA and IPP Set Systems

Note that we can associate the set  $\{(1, x^{(1)}), \dots, (r, x^{(r)})\}$  with the codeword  $(x^{(1)}, \dots, x^{(r)})$  so that the condition for the set system being a c-TA (c-IPP resp.) set system is equivalent to the condition for the code being a c-TA (c-IPP resp.) code. Hence, all c-TA (c-IPP resp.) can form c-TA (c-IPP resp.) set systems, but the converse is not true.

Hence, we can think of c-TA (c-IPP resp.) set systems as generalizations of their corresponding code variants.

## Reed Solomon Based Set Systems

### Theorem (5)

*If  $c \geq 2$ ,  $c \in \mathbb{Z}$  and  $C$  is a Reed-Solomon code of length  $r$  with minimum distance  $d \leq r - \frac{r}{c^2}$ , then the set system corresponding to  $C$  is not a  $c$ -TA set system.*

Choose a codeword  $v \in C$ . Let the corresponding set in the set system be  $v'$ . Let the set system corresponding to  $C$  be  $C'$ . Assume  $C'$  is a  $c$ -TA set system. We will construct a pirate set  $w$  which can be created by the coalition  $C'_0 = \{u'_1, u'_2, \dots, u'_b\}$  with  $b \leq c$  such that  $v' \notin C'_0$  but

$$|v' \cap w| \geq |u'_i \cap w|$$

implying  $v'$  would be caught as a traitor using the tracing traitors algorithm even though he isn't, contradicting the assumption of  $C'$  being a  $c$ -TA set system.

## Reed Solomon Based Set Systems

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Let  $\delta = r - d = k - 1$  (since RS codes are MDS). By assumption, we have  $d \leq r - \frac{r}{c^2} \implies \delta \geq \frac{r}{c^2}$ .

**Case 1:** Assume  $c\delta \leq r$ . For  $i = 1, 2, \dots, c$ , choose  $u_i \in C$ ,  $u_i \neq v$  such that

$$u_i^{(j)} = v^{(j)} \quad \forall j \in \{(i-1)\delta + 1, (i-1)\delta + 2, \dots, i\delta - 1, i\delta\}$$

Note that this is always possible: Let  $f$  be the polynomial that “generates”  $v$  (recall  $v$  is a RS codeword). Let  $h_i$  be a  $\delta$ -degree polynomial which vanishes on the  $\delta$  field elements corresponding to the  $\delta$  positions where  $u_i$  and  $v$  match. Then, we can set  $u_i$  to be the codeword “generated” by the polynomial  $f - h_i$ . Note that this construction automatically ensures  $u_i \neq v$ .

## Reed Solomon Based Set Systems

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Note that by definition of minimum distance, we know that two distinct codewords can agree on at most  $\delta$  positions. Each  $u_i$  matches with  $v$  at exactly  $\delta$  positions. Using the previous construction of  $u_i$ , we can conclude that  $I(u_i, u_j) \leq \delta \forall i \neq j$ . Hence, the number of coordinates where  $u_i$  matches with  $v$  and some  $u_j, j \neq i$  is at most  $\delta + (c - 1)\delta = c\delta$  since there are  $(c - 1)$  choices of  $j \neq i$ .

## Reed Solomon Based Set Systems

### Theorem (5)

*If  $c \geq 2$ ,  $c \in \mathbb{Z}$  and  $C$  is a Reed-Solomon code of length  $r$  with minimum distance  $d \leq r - \frac{r}{c^2}$ , then the set system corresponding to  $C$  is not a  $c$ -TA set system.*

Hence, under the assumption of  $c\delta \leq r$ , we know that each  $u'_i$  contains at least  $(r - c\delta)$  elements that are not in  $v'$  or in  $u'_j$  for any  $j \neq i$ . Since  $(r - c\delta) \geq 0$  and  $c \geq 2$ , we must have

$$r - c\delta \geq \left\lceil \frac{r - c\delta}{c} \right\rceil = \left\lceil \frac{r}{c} \right\rceil - \delta$$

## Reed Solomon Based Set Systems

### Theorem (5)

*If  $c \geq 2$ ,  $c \in \mathbb{Z}$  and  $C$  is a Reed-Solomon code of length  $r$  with minimum distance  $d \leq r - \frac{r}{c^2}$ , then the set system corresponding to  $C$  is not a  $c$ -TA set system.*

To form a pirate set  $w$  from the coalition  $\{u'_1, \dots, u'_c\}$ , first choose the elements  $((i-1)\delta + l, u_i^{(i-1)\delta+l})$  from  $u'_i$  for  $1 \leq i \leq c$  and  $1 \leq l \leq \delta$ .

Next, choose  $\left\lceil \frac{r}{c} \right\rceil - \delta$  elements from each  $u'_i$  that are not in  $v'$  or in  $u'_j$  for any  $j \neq i$ . We can do this since we concluded that the number of such elements are  $\geq \left\lceil \frac{r}{c} \right\rceil - \delta$  and none of them overlap with the elements chosen earlier.

This gives a total of  $c \left\lceil \frac{r}{c} \right\rceil \geq r$  elements for  $w$ . If the number of elements is  $> r$ , just throw away elements until only  $r$  elements are left.

## Reed Solomon Based Set Systems

### Theorem (5)

*If  $c \geq 2$ ,  $c \in \mathbb{Z}$  and  $C$  is a Reed-Solomon code of length  $r$  with minimum distance  $d \leq r - \frac{r}{c^2}$ , then the set system corresponding to  $C$  is not a  $c$ -TA set system.*

For each  $i$ , notice that

$$|u'_i \cap w| \leq \delta + \left( \left\lceil \frac{r}{c} \right\rceil - \delta \right) = \left\lceil \frac{r}{c} \right\rceil$$

$$|v' \cap w| = c\delta \geq \left\lceil \frac{r}{c} \right\rceil$$

Thus, the TA algorithm will mark  $v'$  as a traitor.

**Case 2:** Assume  $c\delta > r$ . In this case, simply choose  $\{u_1, \dots, u_j\}$  as above, where  $j = \left\lfloor \frac{r}{\delta} \right\rfloor < c$ , and choose  $u_{j+1} \neq v$  to agree with  $v$  on only the last  $r - j\delta$  positions. The coalition  $\{u'_1, \dots, u'_j, u'_{j+1}\}$  can create  $v'$  as the pirate set.

## Reed Solomon Based Set Systems

### Theorem (5)

*If  $c \geq 2$ ,  $c \in \mathbb{Z}$  and  $C$  is a Reed-Solomon code of length  $r$  with minimum distance  $d \leq r - \frac{r}{c^2}$ , then the set system corresponding to  $C$  is not a  $c$ -TA set system.*

The consequence of this theorem is that we now have a necessary condition for Reed-Solomon codes to be a  $c$ -TA code. However, we must note that a code may still have TA when the IPP algorithm is used even though the TA algorithm gives incorrect output. In fact, this is posed as an open question by the author. The following theorem gives a family of counter-examples to the previous statement, giving examples of RS codes of length  $r$  and minimum distance  $r - \frac{r}{c^2}$  which are not  $c$ -IPP.

## More Counter-Examples

### Theorem (6)

Let  $s, c \geq 2$  be positive integers,  $p > c^2$  a prime. Define:

- $a_i = (i-1)c$  for  $i = 1, \dots, c$ .
- $g_i(x) = x^s - i$  if  $p \nmid s$ ; else  $g_i(x) = x^s + x - i$ .
- $T =$  roots of  $\{g_i - a_j\}_{1 \leq i, j \leq c}$  inside  $\mathbb{F}_q$  for  $q = p^m$  sufficiently large.

Then  $|T| = c^2s$ , and the RS code  $C$  evaluating degree- $\leq s$  polynomials at  $T$  satisfies:

$$k = s + 1, \quad r = c^2s, \quad d = r - s = r - r/c^2,$$

and  $C$  is **not**  $c$ -IPP.

## More Counter-Examples

We first show that  $T$  consists of  $c^2$ s distinct elements. Let

$h_{ij} = g_i - a_j$ . Then,

$$h_{ij} - h_{mn} = -i - (j-1)c + m + (n-1)c$$

If  $h_{ij}(x) - h_{mn}(x) = 0$ , then  $c|(m-i)$ . However, we know that  $1 \leq m, i \leq c \implies m = i$ .

Hence, we get the relation

$$(j-1)c = (n-1)c \implies j = n$$

Hence, the set  $\{h_{ij}(x)\}$  has unique elements for all  $1 \leq i, j \leq c$ , consisting of  $c^2$  distinct polynomials of degree  $s$ , any two of which differ by a non-zero constant (hence, no common roots).

## More Counter-Examples

The derivative

$$h'_{ij}(x) = \begin{cases} sx^{s-1} & \text{if } p \nmid s \\ 1 & \text{otherwise} \end{cases}$$

Since  $h'_{ij}(x)$  is always relatively prime to  $h_{ij}(x)$ , all the roots of  $h_{ij}(x) = 0$  are simple (no repeated roots). This implies that  $T$  consists of  $c^2$ s distinct elements as desired.

Consider the Reed-Solomon code defined by evaluating polynomials of degree at-most  $s$  at the elements of  $T$ , that is,  $RS(T, c^2s, s + 1)$ .

Consider the disjoint sets of polynomials  $\{a_1, \dots, a_c\}$  and  $\{g_1, \dots, g_c\}$ . Each polynomial maps to a unique codeword (evaluations at the given points). Hence, consider the two coalitions generated by these two sets of polynomials. Note that both these coalitions will be disjoint.

## More Counter-Examples

Both the coalitions can produce the pirate word given as follows:

For each  $\beta \in T$ , there exists a unique  $(i, j)$  such that

$h_{ij}(\beta) = 0 \iff g_i(\beta) = a_j$ . Set the  $\beta^{th}$  coordinate of the codeword as the value  $a_j$ .

Since we have found two disjoint coalitions forming the same pirate, the intersection is null, hence the code is **not** c-IPP.

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## Finding All Possible Coalitions

A coding-theoretic approach can be used to amass additional piracy information like a list of all coalitions capable of creating a given pirate. Such information is useful in two respects:

- It clears all codewords not appearing in any such coalition of involvement in constructing the pirate word.
- It constitutes useful audit information that may help in the prosecution of a traitor later on.

The algorithms require only that the code have minimum distance greater than  $r - r/c^2$ , and are therefore applicable to the codes in Theorems (2,3,4). They are fast when fast list decoding techniques exist.

## Definitions for the Algorithm

### Definition

A subset  $U \subseteq C$  is a minimal  $c$ -coalition for  $w$  if  $|U| \leq c$ ,  $w \in \text{desc}(C)$ , but  $w \notin \text{desc}(V)$  for any proper subset  $V \subset U$ .

Since every coalition of size at most  $c$  that can create  $w$  is obtainable by appending elements to a minimal one, it suffices to find all minimal  $c$ -coalitions.

### Definition

Given a subset  $S \subseteq \{1, \dots, r\}$  of size  $s$ , define the map  $f_S : \mathbb{F}_q^r \longrightarrow \mathbb{F}_q^{r-s}$  to be the projection that omits the coordinates in positions corresponding elements of  $S$ .

## High Level Overview

At a high level, the main algorithm builds a "tree" from which all  $c$ -coalitions capable of constructing a pirate  $w$  can be extracted.

- At the root of the tree lie all codewords known to be in every such coalition.
- The children are candidate codewords for the next member of the coalition.
- Branches are extended till current coalition covers  $w$  (i.e, is capable of constructing it), or until it becomes clear that this is impossible (coalition has size  $c$  but  $w$  cannot be created). Dead-end branches are discarded and other branches are explored.

## Algorithm Sketch (1)

### Input / Output

**Input:** Integer  $c > 1$ , code  $C$  of length  $r$  with minimum distance  $> r - r/c^2$ , pirate word  $w \in \text{desc}_c(C)$ .

**Output:** A list of coalitions of size at most  $c$  that can create  $w$ , including all minimal  $c$ -coalitions for  $w$ .

### Step (i)

Use list decoding to find all codewords  $u_1, \dots, u_a \in C$  ( $a \leq c$ ) within distance  $r - r/c$  of  $w$ .

Let  $S$  be the set of positions (i.e, subset of  $\{1, \dots, r\}$ ) where  $w$  agrees with at least one of  $\{u_1, \dots, u_a\}$ .

Set  $s = |S|$ ,  $r_1 = r - s$ ,  $c_1 = c - a$ ,  $C_1 = f_S(C)$ ,  $w_1 = f_S(w)$ .

If  $r_1 = 0$ , output  $\{u_1, \dots, u_a\}$ .

## Algorithm Sketch (2)

### Step (ii)

Use list decoding to find all codewords  $v_{i1}, \dots, v_{ib_i} \in C_i$  ( $b \leq c_i$ ) within distance  $r_i - r_i/c_i$  of  $w_i$ .

If empty, exit to step (iii).

Otherwise, let  $S_i$  be the subset of  $\{1, \dots, r\}$  on which  $w_i$  agrees with  $v_{ib_i}$ , and set  $s_i = |S_i|$ ,  $r_{i+1} = r_i - s_i$ ,  $c_{i+1} = c_i - 1$ ,  $C_{i+1} = f_{S_i}(C_i)$ , and  $w_{i+1} = f_{S_i}(w_i)$ , and recurse with updated parameters.

## Algorithm Sketch (3)

### Step (iii)

Beginning with  $u_1, \dots, u_a$ , form coalitions by adding (lifts to  $C$  of)  $v_{1b_1}, v_{2b_2}$  and so on, until  $w$  is a descendent of the list.

When this process succeeds or dead-ends (i.e,  $w$  is not a descendent of current list but there are no codewords within distance  $r_i - r_i/c_i$  of  $w_i$  or the list already has  $c$  codewords), move back through the  $v_{ij}$ 's to find the first unexplored branch and repeat step (ii) with a different  $v_{ij}$  in place of  $v_{ib_i}$ .

Terminate when all branches have been explored (all  $v_{ij}$  must be considered).

## Analysis of Algorithm

The output of the algorithm is clearly a list of coalitions of size at most  $c$  that can create the pirate, and includes each minimal  $c$ -coalition at least once.

By Theorem (1) we know that,  $u_1, \dots, u_a$  are in every coalition that create  $w$ .

In Step (ii), if  $d_i > r_i - r_i/c_i^2$  where  $d_i$  is the minimum distance of  $C_i$ , then every coalition that can produce the original pirate  $w$  will contain some list to the original code of some  $v_{ij}$ . Moreover, if a lift to  $C$  of  $v_{ij}$  is in some coalition that can create the original pirate  $w$ , then there exists a codeword within  $r_i - r_i/c_i$  of  $v_{ij}$  (by pigeonhole principle), and the algorithm will proceed.

When  $C$  satisfies any of the sets of conditions in Theorems (2,3,4), then Step (i) can be done efficiently (poly time in  $r$ ).

## Run-time analysis

- **Brute-Force:**

The brute-force method for finding all coalitions will compare for each of the at most  $N^c$  coalitions of size at most  $c$ , each of the  $r$  entries of the pirate to the corresponding entry of each member of the coalition. This gives a time complexity of  $O(crN^c)$ . For Reed-Solomon codes with  $r = \Theta(c^2k)$ , the runtime obtained is  $O(c^3N \log N)$ .

- **Proposed Algorithm:**

In the proposed algorithm we first list decode to find all codewords  $u_1, \dots, u_a$  ( $1 \leq a \leq c$ ) within distance  $r - r/c$  of the pirate, and then use brute force to determine the remaining (at most  $c - a$ ) members. Thus, For Reed-Solomon codes satisfying the conditions of Theorem (2) with  $r = \Theta(c^2k)$ , the runtime is  $O(c^3N^{c-a} \log N)$  which is faster than pure brute force as  $a \geq 1$ .

## Tracing Traitors with Extra Information

Other coding-theoretic techniques may be applied to the traitor tracing problem when additional information about traitor behavior is available. Two main strategies arise depending on what is known:

- **Strategy 1: (Ringleader/Scapegoat)**

If one traitor contributed more than the other members of the coalition to the pirate, apply bounded-distance decoding up to the error-correction bound to find such a traitor very quickly. The run time of bounded-distance decoding for Reed-Solomon codes is at most quadratic in the length of the codewords.

- **Strategy 2: (Roughly equal contributions)**

If all traitors contributed roughly equal amounts, list decoding should be tried first. Traitors can be searched for in sequences of expanding Hamming balls around the pirate, run in parallel or sequentially.

## Soft-Decision Decoding for Traitor Tracing

If information about the traitors or their mode of operation is available, one can build it into a reliability matrix and apply soft-decision decoding algorithms to trace.

Suppose we know that a traitor who contributed the first entry to the pirate also contributed at least  $r/c$  entries, and the underlying code is an RS code, then we can construct a reliability matrix as follows:

**Column 1:** Place a 1 in the entry corresponding to the field element in position 1 of the pirate, and 0's elsewhere.

**Columns  $j > 1$ :** Place  $1 - \epsilon$  in the entry for the field element in position  $j$  of the pirate and all other entries as  $\frac{\epsilon}{q-1}$ , where  $\epsilon < \frac{q-1}{q}$  and is chosen as to optimise the Koetter-Vardy algorithm.

If the entry contributed by the traitor who contributed the most is not known, one method is to choose entries at random from the pirate and apply the above strategy to search for traitors.

## Erasure and Error Decoding

Erasure-and-error decoding may be useful in fingerprinting or watermarking scenarios.

One example of a Erasure model is where, a coalition creates a pirate copy of the digital content by leaving fixed all codeword entries where they all agree, and choosing the values of the remaining positions from  $Q \cup \{?\}$ , where  $Q$  is the alphabet and  $?$ 's can be viewed as erasures.

These techniques are applicable to any scenario in which partial knowledge of the pirate's construction method is available, and complement the list decoding approach described in the previous sections.

## Conclusion

We have demonstrated that traitor tracing algorithms can be quite efficient when the construction of the TA scheme is based on error-correcting codes and the method of tracing is based on fast list decoding algorithms.

For the TA algorithm, traitors can be identified in poly time in  $r$ , where  $r$  is roughly  $O(c^2 \log N)$ , rather than in  $O(N)$  time. We also show that the same techniques can be used to build an algorithm for finding all possible traitor coalitions. Over time, this algorithm can be used to accumulate convincing evidence against additional users. We also give evidence for a close relationship between the TA and IPP properties for linear codes.

Finally, we also see avenues for future research, including applications of soft-decision and erasure decoding techniques to traitor tracing in specific scenarios.